$$\underline{r} = \begin{pmatrix} 2+\lambda+\mu \\ 2+\lambda-5\mu \\ 1-2\mu \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2+\lambda+\mu \\ 2+\lambda-5\mu \\ 2-2\mu \end{pmatrix} \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

$(1)-(2)$: $6\mu = x-y \Rightarrow \mu = \frac{x-y}{6}$

$(3)-2(1)$: $2+0-2\mu - (4+2\lambda-2\mu)$

$z-2x = -2-2\lambda$

$2\lambda = 2x-z-2$

$x = 2+\left(x-\frac{z}{2}-1\right)+\frac{x-y}{6}$

$6x = 12+6x-3z-6+x-y$

$x-y-3z+6=0$

$$\underline{r}\cdot\begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}\cdot\begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix} = -3$$

$$\underline{r}\cdot\begin{pmatrix} 7 \\ 17 \\ -3 \end{pmatrix}$$

$(0,0,7)$

$$\frac{-3(7)-6}{\sqrt{3^2}} = 3$$

$$\cos\theta = \frac{|7-17+9|}{\sqrt{347}\sqrt{11}} = \frac{1}{\sqrt{11}\sqrt{347}}$$

$\theta = 89.1$

$$(4\sinh x + 3\cosh x)^2 = 16\sinh^2 x + 24\sinh x\cosh x + 9\cosh^2 x$$

$$16\left(\tfrac{1}{2}\cosh 2x - \tfrac{1}{2}\right) + 12\sinh 2x + 9\left(\tfrac{1}{2} + \tfrac{1}{2}\cosh 2x\right)$$

$$\left(8 + \tfrac{9}{2}\right)\cosh 2x + 12\sinh 2x + \tfrac{9}{2} - 8$$

$$\tfrac{25}{2}\cosh 2x + 12\sinh 2x - \tfrac{7}{2}$$

$$\pi\int \tfrac{25}{2}\cosh 2x + 12\sinh 2x - \tfrac{7}{2}\,dx$$

$$\pi\left[\tfrac{25}{4}\sinh 2x + 6\cosh 2x - \tfrac{7}{2}x\right]_{\frac{1}{2}\ln\frac{1}{7}}^{0}$$

$$4\sinh x = -3\cosh x$$
$$4\tanh x = -3$$
$$\tanh x = -\tfrac{3}{4}$$

$$\frac{e^x - e^{-x}}{e^x + e^{-x}} = -\tfrac{3}{4}$$

$$-3(e^x + e^{-x}) = 4(e^x - e^{-x})$$
$$-3e^{2x} - 3 = 4e^{2x} - 4$$
$$7e^{2x} - 1 = 0$$
$$e^{2x} = \tfrac{1}{7}$$
$$\ln e^{2x} = \ln\tfrac{1}{7}$$
$$x = \tfrac{1}{2}\ln\tfrac{1}{7}$$

$$\sin^2 x = \tfrac{1}{2} - \tfrac{1}{2}\cos 2x$$
$$-\sinh^2 x = \tfrac{1}{2} - \tfrac{1}{2}\cos 2x$$
$$\sinh^2 x = \tfrac{1}{2}\cos 2x - \tfrac{1}{2}$$
$$\cosh^2 x = \tfrac{1}{2} + \tfrac{1}{2}\cos 2x$$
$$2\sinh x\cosh x = \sinh 2x$$

$$\pi\left[\frac{25}{4}\sinh 2x + 6\cosh 2x - \frac{7}{2}x\right]_{\frac{1}{2}\ln\frac{1}{7}}^{0}$$

$$\left(0 + 6 - 0\right) - \left(\frac{25}{4}\sinh\ln\left(\tfrac{1}{7}\right) + 6\cosh\ln\tfrac{1}{7} - \tfrac{7}{4}\ln\tfrac{1}{7}\right)$$

$$\frac{e^{\ln\frac{1}{7}} - e^{-\ln\frac{1}{7}}}{2} = \frac{\frac{1}{7} - 7}{2} \qquad \frac{e^{\ln\frac{1}{7}} + e^{-\ln 7}}{2} = \frac{\frac{1}{7} + 7}{2}$$

$$\frac{\frac{25}{7} - 175}{8} + \frac{\frac{6}{7} + 42}{2} + \frac{7}{4}\ln\tfrac{1}{7}$$

$$6 + \tfrac{7}{4}\ln\tfrac{1}{7} \qquad 6 - \tfrac{7}{4}\ln 7$$

$f(x) = ax^2 + bx + c$

$f'(x) = 2ax + b$

$2a(5) + b = 0$
$10a + b = 0$

$f(4) = a(4^2) + b(4) + c = 0$
$16a + 4b + c = 0$

$(5, 4)$ $\quad 4 = a(5^2) + b(5) + c$
$25a + 5b + c = 4$

$-10a = b$

$16a + 4(-10a) + c = 0$
$25a + 5(-10a) + c = 4$

$-24a + c = 0$
$-25a + c = 4$

$a = -4$
$-10(-4) = 40 \quad b = 40$

$-25(-4) + c = 4$
$100 + c = 4$
$c = -96$

Series Solution

$$\frac{d^2y}{dx^2} + y\frac{dy}{dx} - 4y = 0$$

$x_0 = 0 \quad y_0 = 3 \quad \left.\frac{dy}{dx}\right|_{x=0} = 2$

$$\frac{d^2y}{dx^2} = 4y - y\frac{dy}{dx} \qquad \left.\frac{d^2y}{dx^2}\right|_{x=0} = 6$$

$$\frac{d^3y}{dx^2} + y\frac{d^2y}{dx^2} + \frac{dy}{dx}\frac{dy}{dx} - 4\frac{dy}{dx} = 0$$

$$\left.\frac{d^3y}{dx^3}\right|_{x=0} = 4(2) - (2)(2) - 3(6) = 8 - 4 - 18 = -14$$

$$y = 3 + 2x + 6\frac{x^2}{2!} + -14\frac{x^3}{3!}$$

$$= 3 + 2x + 3x^2 - \frac{7}{3}x^3$$

$$\sin\theta = \frac{a \cdot b}{|a||b|} = \frac{15 + 0 + 12}{\sqrt{34}\sqrt{34}} \qquad \underline{a} = \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} \quad \underline{b} = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix}$$

$$\sin\theta = \frac{27}{34} \Rightarrow \theta =$$

$$\sqrt{3^2 + 3^2 + 4^2} = \sqrt{34}$$

$$\sqrt{5^2 + 3^2} = \sqrt{34}$$

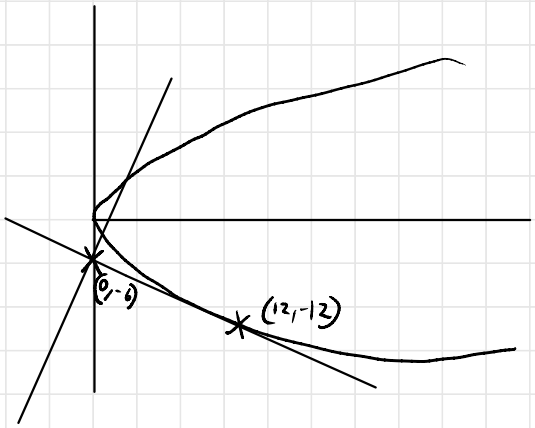
$$4\begin{vmatrix} x & 0 \\ 4 & 2x \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 3 & 2x \end{vmatrix} - x\begin{vmatrix} 1 & x \\ 3 & 4 \end{vmatrix}$$

$$= 4(2x^2) - (2x) - x(4 - 3x)$$

$$8x^2 - 2x - 4x + 3x^2$$

$$11x^2 - 6x = 0 \qquad x(11x - 6) = 0 \qquad x = \frac{6}{11}$$

$y^2 = 12x \qquad a = 3$



$2y\frac{dy}{dx} = 12$

$\frac{dy}{dx} = \frac{12}{2y} = \frac{6}{y}$

$\frac{dy}{dx} = \frac{6}{-12} = -\frac{1}{2}$

$y + 12 = -\frac{1}{2}(x - 12)$

$2y + 24 = -x + 12$

$x + 2y + 12 = 0$

$2y + 12 = 0 \Rightarrow y = -6$

$2y = -x - 12$

$y = -\frac{1}{2}x - 6$

$y = 2x + k$

$-6 = 2(0) + k \Rightarrow k = -6$

$$\begin{pmatrix} 2\lambda+1 \\ 3\lambda+2 \\ 4\lambda+3 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \lambda$$

$$r = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \lambda + \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix} \mu$$